

Title (英文) : The Inverse Law of Local Time-Temperature within the Angular-Momentum Network Geometry (ANG) Framework

标题 (中文) : 角动量网络几何学 (ANG) 框架下的时间-温度反比律

Subtitle (英文) : Topological Coupling Between Local Time Flow Rate and Temperature

副标题 (中文) : 局域时间流速与温度的拓扑耦合

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Abstract 摘要

English

General relativity establishes the relationship between gravitational potential and local time flow rate (gravitational time dilation). Within the Angular-Momentum Network Geometry (ANG) framework, this paper demonstrates that there exists a rigid inverse proportional relationship between the local time flow rate and temperature. This relation originates from two core axioms of ANG: Theorem T3 (Spectral Isomorphism Theorem), which defines temperature as the first-order moment of the energy spectrum of link networks, and Axiom XI, which defines the local time step as the reciprocal of the topological reconnection rate.

We derive the closed-form relation:

$$\frac{dt_{\text{local}}}{dt_{\text{ref}}} = \frac{T_{\text{ref}}}{T_{\text{local}}}$$

This effect is comparable in magnitude to black-body radiation (BBR) corrections, yet it exhibits distinct scaling behavior and element-dependent responses. ANG predicts that two atomic clocks of different elements (e.g., Cs and Yb) will show systematic differences in their nonlinear frequency-drift rates under identical temperature gradients, directly correlated with their topological parameters (χ_{Link}).

Keywords: time-temperature inverse law; Angular-Momentum Network Geometry; spectral isomorphism; atomic-clock drift; topological time

中文

广义相对论建立了引力势与局域时间流速之间的联系（引力时间膨胀）。本文在角动量网络几何学（ANG）框架内证明：局域时间流速与温度之间存在刚性反比关系。这

一关系源自 ANG 的两个核心公理：T3（谱同构定理）将温度定义为链接网络能量谱的一阶矩，Axiom XI 将时间步长定义为重联速率的倒数。

我们推导出闭式关系：

$$\frac{dt_{\text{local}}}{dt_{\text{ref}}} = \frac{T_{\text{ref}}}{T_{\text{local}}}$$

该效应在量级上与黑体辐射（BBR）修正可比，但标度行为、原子种类依赖性完全不同。ANG 预言：两种不同元素原子钟（例如铯 Cs 与镱 Yb）在相同温度梯度下，非线性频率漂移率将出现系统性差异，该差异与原子拓扑参数 (χ, Link) 直接相关。

关键词： 时间-温度反比律；角动量网络几何学；谱同构；原子钟漂移；拓扑时间

1. Introduction 引言

English

In modern physics, the flow rate of local time is known to be modified by two factors:

1. Gravitational potential (General Relativity): Clocks run slower in stronger gravitational fields.
2. Relative velocity (Special Relativity): Moving clocks run slower.

This paper proposes a third fundamental regulator: local temperature.

Within the standard physical framework, temperature-induced shifts in atomic-clock frequency are interpreted as the black-body-radiation (BBR) Stark effect — the thermal radiation field shifts atomic energy levels. This is an electromagnetic interpretation rather than a geometric one. From the ANG perspective, temperature modifies time via changes in the spectral density of the underlying link network, not via electromagnetic radiative perturbation.

中文

在现代物理学中，已知有两种因素可以改变局域时间流速：

1. 引力势（广义相对论）：引力场越强，时钟走时越慢。
2. 相对速度（狭义相对论）：高速运动的时钟走时变慢。

本文提出第三个底层调控因素：局域温度。

在标准物理框架中，温度造成原子钟频率偏移被解释为黑体辐射（BBR）斯塔克效应——热辐射场扰动原子能级。这是电磁层面的解释，而非几何本源解释。在 ANG 视角下，温度改变时间流速的根源是底层链接网络谱密度发生改变，并非来自电磁辐射扰动。

2. Definition of Time in the ANG Framework ANG 框架中的时间定义

2.1 Time = Reconnection Rate 时间 = 重联速率

English

According to Axiom XI, the local discrete time step is given by:

$$\Delta t_{\text{local}} = \frac{\bar{\ell}}{\bar{\kappa} \cdot \omega_{\text{topo}}}$$

Where:

- $\bar{\ell}$: average topological spacing of the link network
- $\bar{\kappa}$: local angular stiffness
- ω_{topo} : universal intrinsic topological reconnection frequency (global constant)

中文

根据 Axiom XI, 局域离散时间步长满足:

$$\Delta t_{\text{local}} = \frac{\bar{\ell}}{\bar{\kappa} \cdot \omega_{\text{topo}}}$$

符号说明:

- $\bar{\ell}$: 链接网络平均拓扑间距
- $\bar{\kappa}$: 局域角刚度
- ω_{topo} : 宇宙本征拓扑重联频率 (全局常数)

2.2 Temperature as the First-Order Spectral Moment 温度 = 谱密度一阶矩

English

From Theorem T3 (Angular-Momentum Spectral Isomorphism Theorem), temperature is defined as the normalized first-order moment of the link-network energy spectrum:

$$T = \frac{1}{k_B} \cdot \frac{\int_0^\infty \lambda \mathcal{S}(\lambda), d\lambda}{\int_0^\infty \mathcal{S}(\lambda), d\lambda}$$

Temperature is proportional to the ratio between the first-order moment and the zero-order moment of the spectral density $\mathcal{S}(\lambda)$.

中文

依据 T3 (角动量-谱同构定理), 温度被定义为链接网络能谱归一化一阶矩:

$$T = \frac{1}{k_B} \cdot \frac{\int_0^\infty \lambda \mathcal{S}(\lambda), d\lambda}{\int_0^\infty \mathcal{S}(\lambda), d\lambda}$$

温度正比于能量谱密度 $\mathcal{S}(\lambda)$ 的一阶矩与零阶矩的比值。

3. Coupling Between Temperature and Angular-Stiffness 温度-角刚度耦合

English

In ANG, angular stiffness $\bar{\kappa}$ is directly related to the second-order moment of spectral density (stiffness corresponds to the curvature of the potential energy, i.e., spectral width):

$$\bar{\kappa} \propto \frac{\int_0^\infty \lambda^2 \mathcal{S}(\lambda), d\lambda}{\int_0^\infty \mathcal{S}(\lambda), d\lambda}$$

As temperature rises:

1. High-frequency tails of spectral density $\mathcal{S}(\lambda)$ are excited;
2. The second-order moment (spectral width) increases $\rightarrow \bar{\kappa}$ rises;
3. The first-order moment (average energy) increases $\rightarrow T$ rises.

Before anharmonic saturation occurs, we obtain the proportionality $\bar{\kappa} \propto T$.

Substituting into the time-step formula yields:

$$\Delta t_{\text{local}} \propto \frac{1}{T}$$

中文

在 ANG 体系中，角刚度 $\bar{\kappa}$ 和谱密度的二阶矩直接关联（刚度代表势能曲率，对应能谱宽度）：

$$\bar{\kappa} \propto \frac{\int_0^\infty \lambda^2 \mathcal{S}(\lambda), d\lambda}{\int_0^\infty \mathcal{S}(\lambda), d\lambda}$$

当温度升高：

1. 谱密度 $\mathcal{S}(\lambda)$ 的高频尾部被激发；
2. 二阶矩（谱宽）增大 $\rightarrow \bar{\kappa}$ 增大；
3. 一阶矩（平均能量）增大 $\rightarrow T$ 升高。

在非谐饱和效应出现之前，满足比例关系 $\bar{\kappa} \propto T$ ；代入时间步长公式得到：

$$\Delta t_{\text{local}} \propto \frac{1}{T}$$

4. The Time-Temperature Inverse Law 时间-温度反比律

English

$$\frac{dt_{\text{local}}}{dt_{\text{ref}}} = \frac{T_{\text{ref}}}{T_{\text{local}}}$$

Where dt_{ref} denotes time flow at reference temperature T_{ref} , and dt_{local} denotes local time flow at temperature T_{local} .

Condition	Reconnection Rate	Time Step	Time Flow Rate
High temperature	Accelerated	Shortened	Faster
Low temperature	Decelerated	Lengthened	Slower

中文

$$\frac{dt_{\text{local}}}{dt_{\text{ref}}} = \frac{T_{\text{ref}}}{T_{\text{local}}}$$

式中 dt_{ref} 为参考温度 T_{ref} 下的时间流速； dt_{local} 是局域温度 T_{local} 下的时间流速。

条件	重联速率	时间步长	时间流速
高温	加快	缩短	变快
低温	减慢	延长	变慢

5. Comparison with Gravitational Time Dilation (General Relativity) 与广义相对论引力时间膨胀对比

English

Dimension	Gravitational Time Dilation (GR)	ANG Time-Temperature Inverse Law
Driving source	Gravitational potential gradient	Topological-density gradient
Trend	Stronger field → time runs slower	Higher temperature → time runs faster

Species dependence	None (all clocks behave identically)	Dependent on atomic topological parameters χ, Link
Experimental window	Difficult laboratory tests	Cross-comparison of multi-element atomic clocks

中文

维度	引力时间膨胀（广义相对论 GR）	ANG 时间-温度反比律
驱动力	引力势梯度	拓扑密度梯度
变化方向	强引力场→时间变慢	高温→时间变快
物质种类依赖性	无（所有时钟变化一致）	与原子拓扑参数 χ, Link 相关，存在元素差异
可实验窗口	实验室条件下检验难度高	多元素原子钟交叉比对可验证

6. Falsifiable Predictions 可证伪预测

6.1 Nonlinear Drift of Multi-Element Atomic Clocks 不同元素原子钟非线性漂移

English

Within the standard BBR framework, frequency-drift differences among different atomic species are mainly determined by electric-dipole moments. ANG predicts an additional topological correction term related to atomic topological invariants:

$$\frac{\Delta v_{\text{topo}}}{v_0} = -\alpha \cdot \frac{\Delta T}{T} \cdot \mathcal{F}(\chi, \text{Link})$$

$\mathcal{F}(\chi, \text{Link})$ is the topological factor determined by the link-network structure of the corresponding atom.

Direct evidence for ANG would be observed if Cs clocks and Yb clocks exhibit systematic deviations proportional to $\mathcal{F}(\chi, \text{Link})$ under identical temperature gradients.

中文

在标准黑体辐射 BBR 理论中，不同原子频率漂移差异主要由电偶极矩决定。ANG 预言存在额外一项由原子拓扑不变量决定的拓扑修正项：

$$\frac{\Delta\nu_{\text{topo}}}{\nu_0} = -\alpha \cdot \frac{\Delta T}{T} \cdot \mathcal{F}(\chi, \text{Link})$$

$\mathcal{F}(\chi, \text{Link})$ 为对应原子链接网络结构给出的拓扑因子。

如果 Cs、Yb 原子钟在完全相同的温度梯度下，出现和拓扑因子 $\mathcal{F}(\chi, \text{Link})$ 成比例的系统性偏差，则可以得到 ANG 理论的直接实验证据。

6.2 Time-Anomaly Near High-Temperature Superconducting Transition 高温超导转变附近的时间异常

English

Near the critical temperature of high-temperature superconductors (at the topological threshold $\Phi \rightarrow 0.85$), ANG predicts a measurable anomaly in local time flow: atomic-clock frequencies will exhibit systematic "freezing" or acceleration offsets near T_c . This unique prediction cannot be explained by conventional BBR corrections.

中文

在高温超导体转变温度附近（拓扑临界点 $\Phi \rightarrow 0.85$ ），ANG 预言局域时间流速会出现可测量异常：原子钟频率在临界温度 T_c 附近产生系统性“冻结”或者加速偏移。该独有的预言无法用传统黑体辐射 BBR 效应解释。

6.3 Clock Comparison: Solar Surface vs. Earth 太阳表面-地球原子钟比对

English

Comparing a clock at the solar surface ($T = 5800\text{K}$) with a terrestrial reference clock ($T = 300\text{K}$):

$$\frac{dt_{\text{Sun}}}{dt_{\text{Earth}}} \approx \frac{300}{5800} \approx 5.2 \times 10^{-2}$$

After subtracting the gravitational time-dilation effect from general relativity, the residual deviation is on the order of 10^{-2} . Separability between the GR term and the ANG temperature term depends on the projection relationship between ANG axiom system and general relativity. This high-precision inter-space clock comparison has not yet been implemented.

中文

将太阳表面（5800 K）原子钟与地球参考时钟（300 K）比对：

$$\frac{dt_{\text{太阳}}}{dt_{\text{地球}}} \approx \frac{300}{5800} \approx 5.2 \times 10^{-2}$$

扣除广义相对论引力时间膨胀效应之后，剩余偏差量级约 10^{-2} 。引力项与 ANG 温度

项能否分离，取决于 ANG 公理体系和广义相对论之间的投影映射关系。该高精度跨天体时钟比对实验目前尚未开展。

7. Conclusions 结论

English

Within the Angular-Momentum Network Geometry (ANG) framework, we derive the rigid inverse relation:

$$\frac{dt_{\text{local}}}{dt_{\text{ref}}} = \frac{T_{\text{ref}}}{T_{\text{local}}}$$

Local time flow rate is inversely proportional to local temperature. Higher temperature accelerates topological reconnection, making local time run faster; lower temperature slows reconnection and makes local time run slower.

This is not a relativistic effect. It is the direct spectral-density projection of link-network reconnection dynamics. Higher temperature enriches high-frequency components in the spectral density, increases the angular stiffness of links, accelerates topological reconnection, and shortens the elementary time step.

中文

在角动量网络几何学（ANG）框架内，我们推导出刚性反比关系：

$$\frac{dt_{\text{local}}}{dt_{\text{ref}}} = \frac{T_{\text{ref}}}{T_{\text{local}}}$$

局域时间流速与温度呈严格反比。温度越高，拓扑重联过程越快，局域时间变快；温度越低，重联被抑制，时间流速变慢。

该效应不属于相对论效应，是链接网络重联动力学在谱密度层面的直接投影：温度升高会激发谱密度高频分量，增大链接角刚度，加速拓扑重联，使基础时间步长缩短。

References 参考文献

English

- [1] Song, C. (2026). ANG-Math v1.2: Mathematical Formalism of Angular-Momentum Network Geometry.
- [2] Song, C. (2026). UVMM-ANG v5.7: Full-Discipline AI Knowledge Migration Package.
- [3] CODATA. (2022). Recommended Values of the Fundamental Physical Constants.
- [4] BIPM. (2024). Recommended Uncertainties for Standard Atomic Clocks.

中文

- [1] 宋承斌. (2026). ANG-Math v1.2: 角动量网络几何学数学体系.
- [2] 宋承斌. (2026). UVMM-ANG v5.7: 全学科 AI 知识迁移包.

- [3] CODATA (2022). 基础物理常数推荐值.
- [4] BIPM (2024). 标准原子钟不确定度推荐规范.